

Commissioning of the Electron Accelerator LEReC for Bunched Beam Cooling

Dmitry Kayran for LEReC team,
Brookhaven National Laboratory

NAPAC'2019
September, 1-6 2019
Lancing, Michigan

BROOKHAVEN
NATIONAL LABORATORY



Outline

- LEReC goals
- Layout
- Beam requirements
- Stages of commissioning
- Critical components
- Beam quality measurements
- High current operation
- First Cooling demonstration
- Summary

LEReC Project Goals

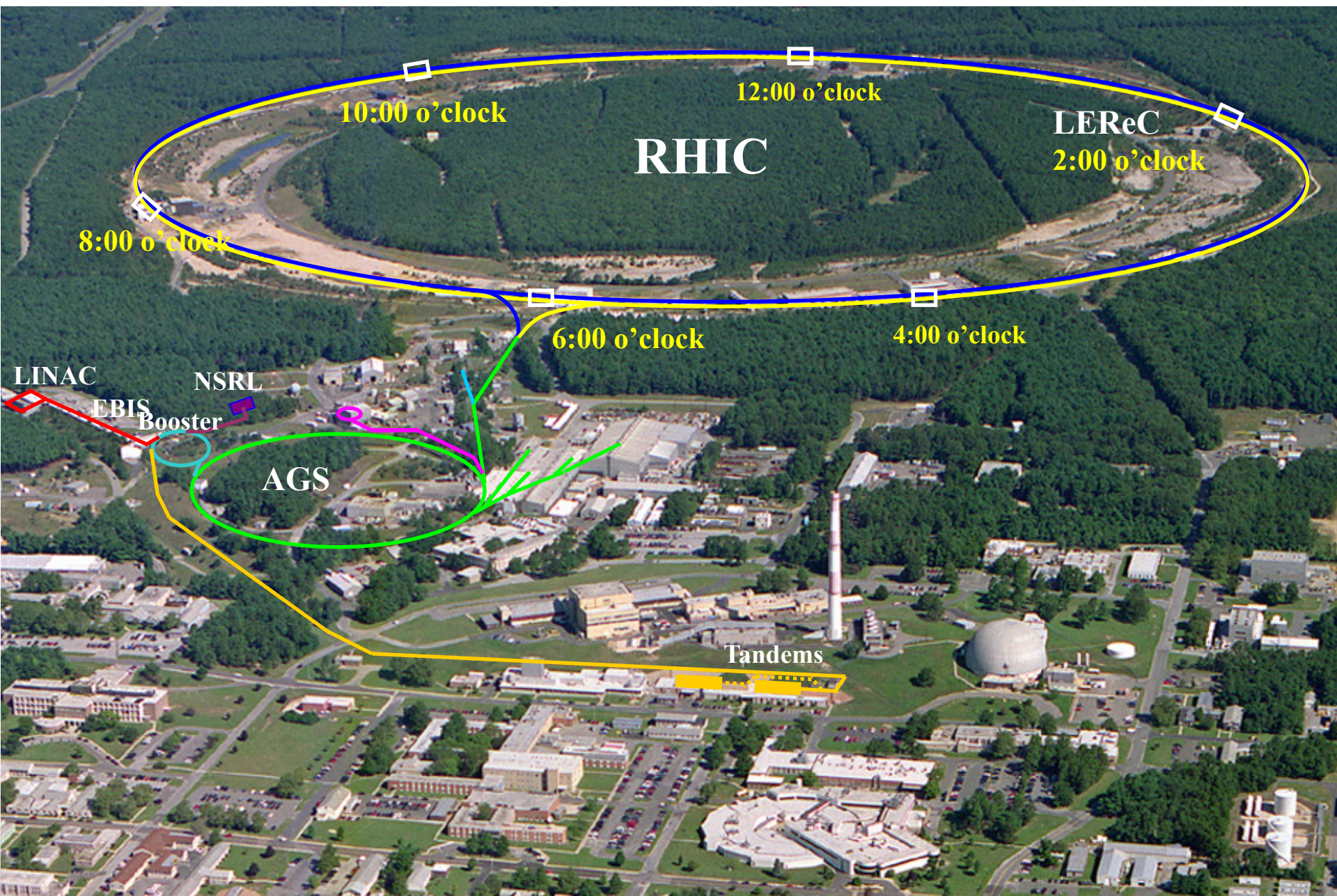
The goal of the LEReC project is to provide luminosity improvement for RHIC operation at low energies to search for a QCD critical point: Beam Energy Scan Phase-II physics program.

LEReC is first RF linac-based electron cooler (bunched beam cooling).

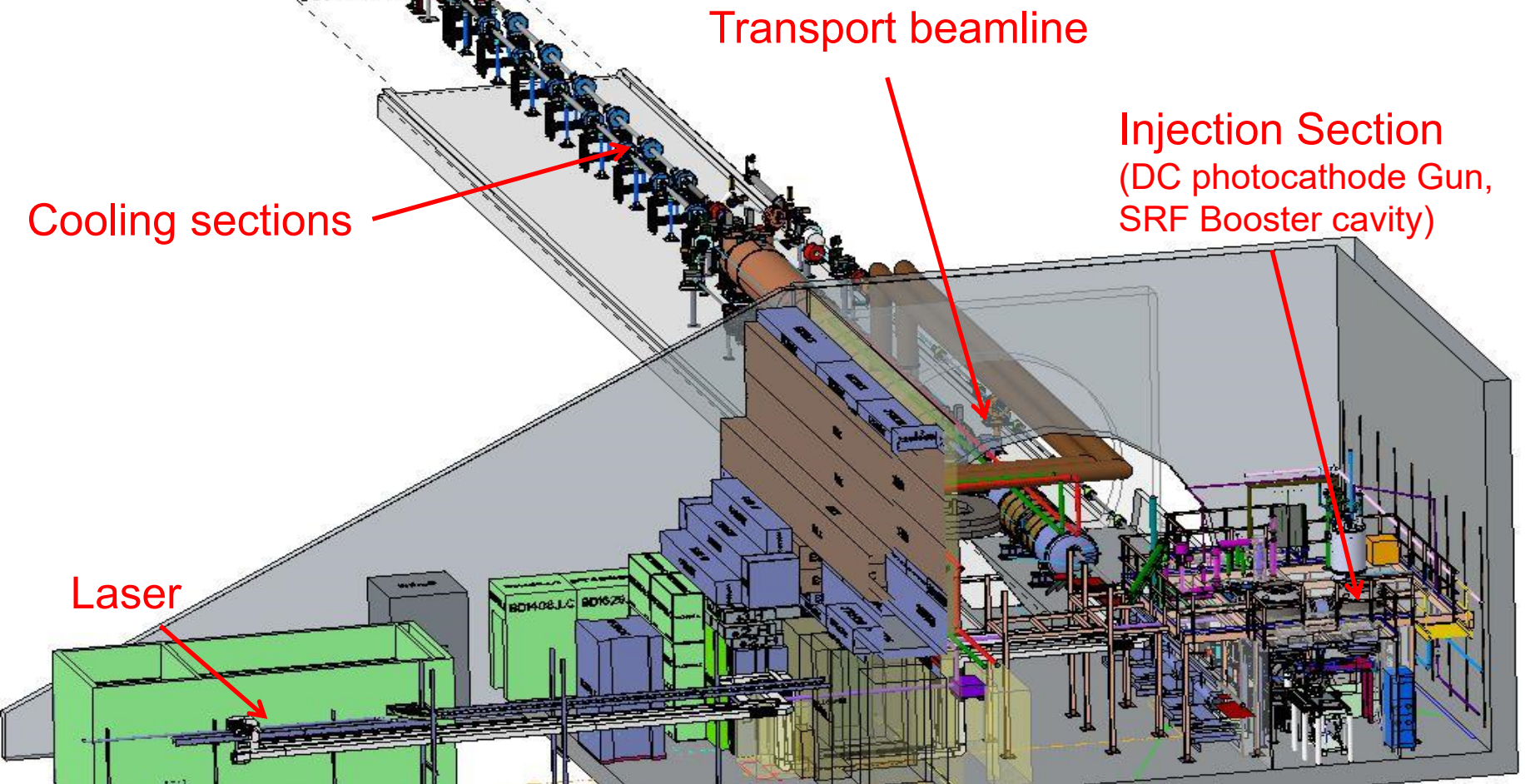
To provide luminosity improvement with such approach requires:

- ❑ Building and commissioning of new state of the art electron accelerator
- ❑ Produce electron beam with beam quality suitable for cooling
- ❑ RF acceleration and transport maintaining required beam quality
- ❑ Achieve required beam parameters in cooling sections
- ❑ Commissioning of bunched beam electron cooling
- ❑ Commissioning of electron cooling in a collider

RHIC @ BNL, Long Island, New York



LEReC inside RHIC tunnel at Interaction Region @ 2 o'clock (IR2)



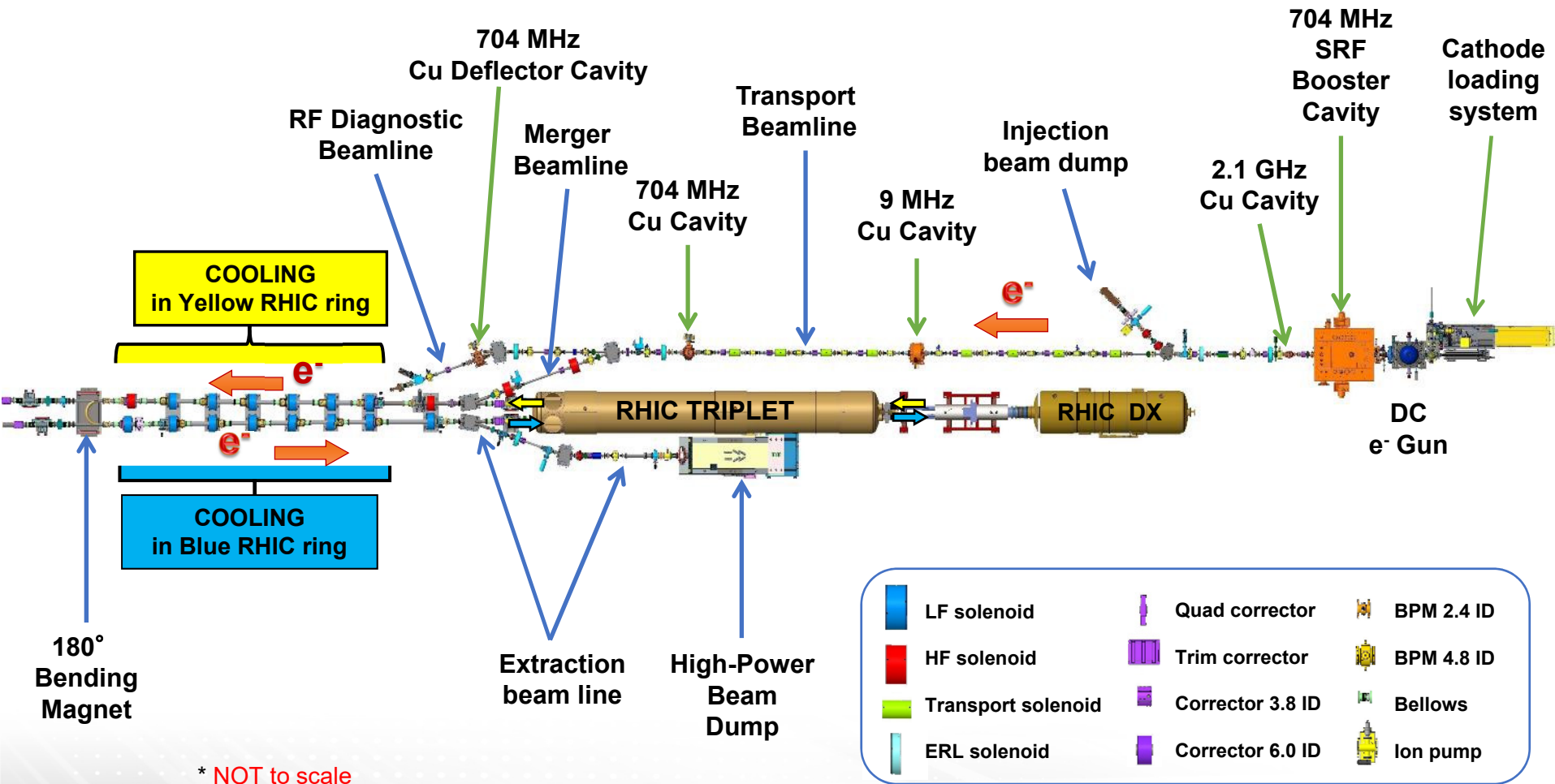
Laser, RF amplifiers, Magnets power supplies,
Beam instrumentation racks are located outside of the RHIC tunnel

Bunched beam electron cooling for LEReC

- Produce electron bunches suitable for cooling by illuminating a multi-alkali photocathode inside the Gun with green light using high-power laser (high-brightness in 3D: both emittance and energy spread).
- The 704 MHz fiber laser produces required modulations to overlap ion bunches at 9 MHz frequency with laser pulse temporal profile shaping using crystal stacking.
- Accelerate such bunches with RF and use RF gymnastics (several RF cavities) to achieve energy spread required for cooling. Deliver and maintain beam quality in both cooling sections.
- Electron bunch overlaps only small portion of ion bunch. All amplitudes are being cooled as a result of synchrotron oscillations.

LEReC Accelerator Layout

(100 meters of beamlines with the DC Gun, high-power fiber laser, 5 RF systems, including one SRF, many magnets and beam instrumentation)



LEReC beam structure used in cooling section

Ions structure:

120 bunches

$f_{\text{rep}} = 120 \times 75.8347 \text{ kHz} = 9.1 \text{ MHz}$

$N_{\text{ion}} = 6e8$, $I_{\text{peak}} = 0.25 \text{ A}$

Rms length = 12 nsec

$N_{\text{ion}} = 6e8$

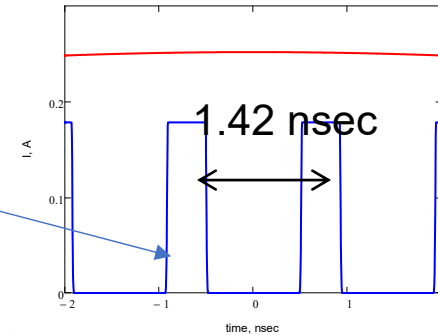
Electron bunches:

$f_{\text{SRF}} = 703.5 \text{ MHz}$

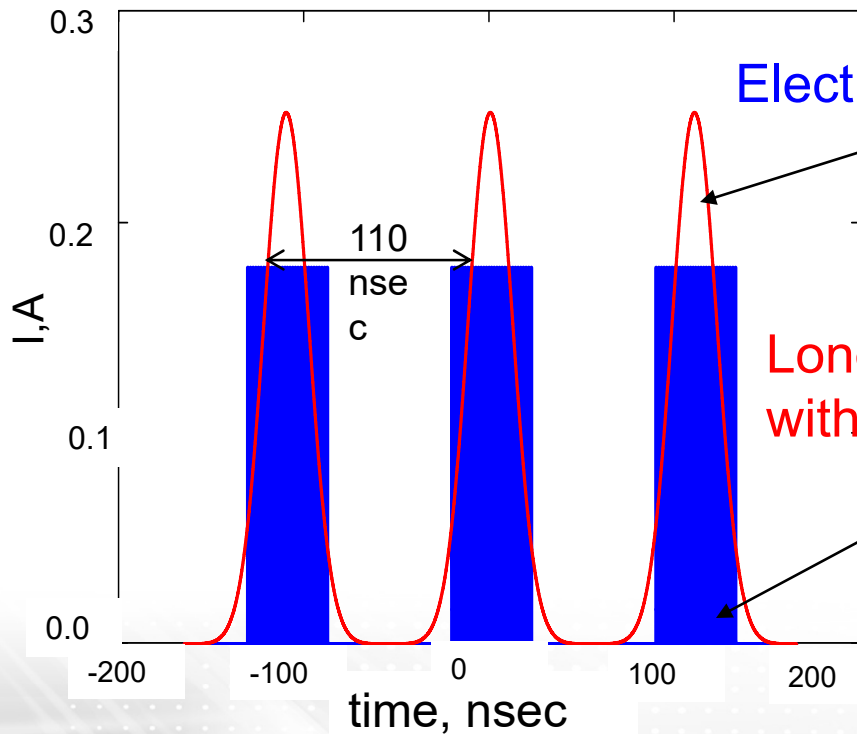
$Q_e = 75 \text{ pC}$, $I_{\text{peak}} = 0.18 \text{ A}$

FWHM = 400 psec

Electron Beam profile



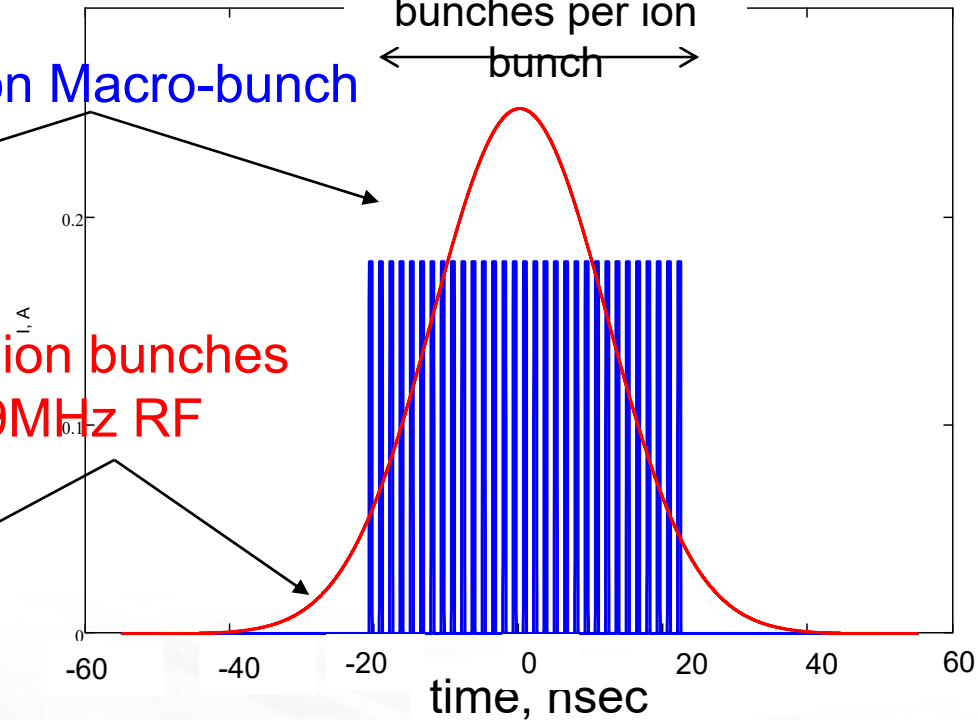
9 MHz bunch structure



30 electron bunches per ion

← bunch →

Long ion bunches with 9 MHz RF



LEReC electron beam parameters

Electron beam requirement for cooling				Measured*
Kinetic energy, MeV	1.6	2	2.6	1.6, 2.0
Pulse duration at the cathode, psec	40	40	40	40
Bunch duration in cool. sect, psec	400	400	20	400
Electron bunch (704MHz) charge, pC	130	170	200	50-200
Bunches per macrobunch (9 MHz)	30	30	24-30	30
Charge in macrobunch, nC	4	5	5-6	1-6
RMS normalized emittance, μm	< 2.5	< 2.5	< 2.5	1.6-2
Average current, mA	36	47	45-55	14-30
RMS energy spread	< 5e-4	< 5e-4	< 5e-4	<2e-4
RMS angular spread in cooling section	<150 urad	<150 urad	<150 urad	100-200 urad



Commissioned
during 2019 RUN

*) not at the
same time

Stages of LEReC Commissioning

- **Phase 1: DC Gun tests**

(April-August 2017):
configuration

DC Gun tests in temporary

(January-February 2018): DC Gun tests in final configuration

- **Phase 2 (March-September 2018): Full LEReC commissioning**

Goals: Meet all required KPPs. Achieve high-current operation of accelerator. Achieve electron beam parameters suitable for cooling.

- **Phase 3 (2018-2019): Transition to operations**

Goals: Achieve required stability (energy, orbit) of electron beam. Develop necessary stability, ripple, intensity, orbit feedbacks. Commissioning of LEReC for operation at higher energies.

Phase 4 (2019-2020): Commissioning of cooling – requires Au ions at the same energy

LEReC timeline

May 2015:	LEReC project approved by DOE for construction
December 2016:	DC gun successfully conditioned in RHIC IR2
February 2017:	Gun Test beamline installed in RHIC
April-Aug., 2017:	First Gun tests with beam
July-Dec., 2017:	Installation of full LEReC accelerator
Jan.-Feb., 2018:	Systems commissioning (RF, SRF, Cryogenics, Instrumentation, Controls, etc.)
March-Sept. 2018:	Commissioning of full LEReC accelerator with e-beam
Sept 2018 :	All project Key Performance Parameters achieved
Oct.-Dec., 2018:	Scheduled upgrades and modifications (NO beam testing)
December, 2018:	Gun Conditioning and HVPS tests without e-beam
Jan.-Feb., 2019:	Restart operation with electron beam. Achieved e-beam quality suitable for cooling
March 2019:	<u>Start commissioning with Au ion beams.</u> Matched RHIC/LEReC beams <u>energies</u> and trajectories in cooling sections
Apr 2019:	First cooling in one ring then in both RHIC rings using 76 kHz bunch
May 2019:	Cooling all bunches in both RHIC rings using 9MHz CW LEReC
June-July 2019:	Cooling optimization, studies effect to RHIC collision rate, Commissioning LEReC at higher energies.

Achieving of “cold” electron beam quality suitable for cooling

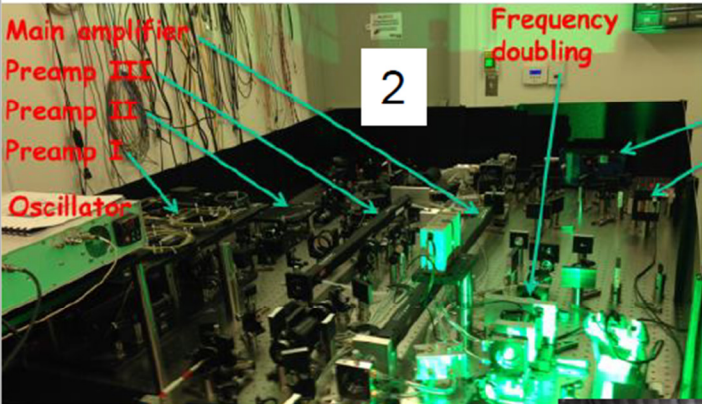
- LEReC is based on the state-of-the-art accelerator physics and technology:
 - Photocathodes: production and delivery system
 - High power fiber laser and transport
 - Laser beam shaping to produce electron bunches of required quality
 - Operation of DC gun at high voltages (around 400kV) with high charge and high average current
 - RF gymnastics using several RF cavities and stability control
 - Energy stability and control
 - Instrumentation and controls

LEReC Critical Technical Systems

1. DC photocathode electron gun and HV PS
2. High-power fiber laser system and transport
3. Cathode production deposition and delivery systems
4. SRF Booster cavity
5. 2.1 GHz and 704 MHz warm RF cavities



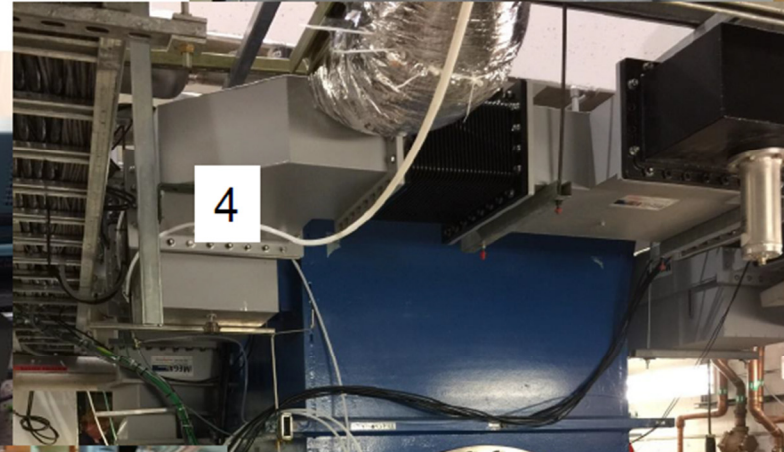
1



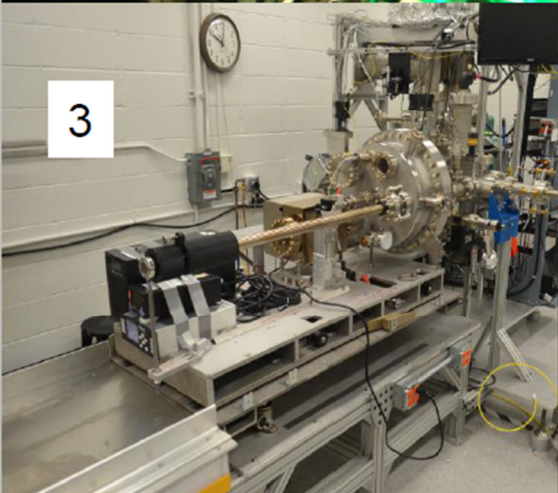
2



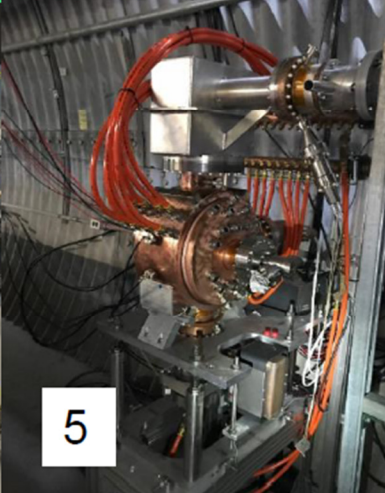
Diagnostic tools



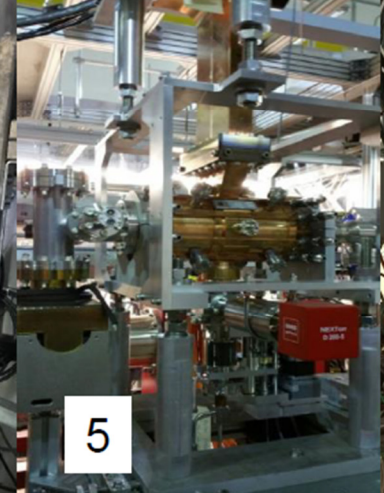
4



3



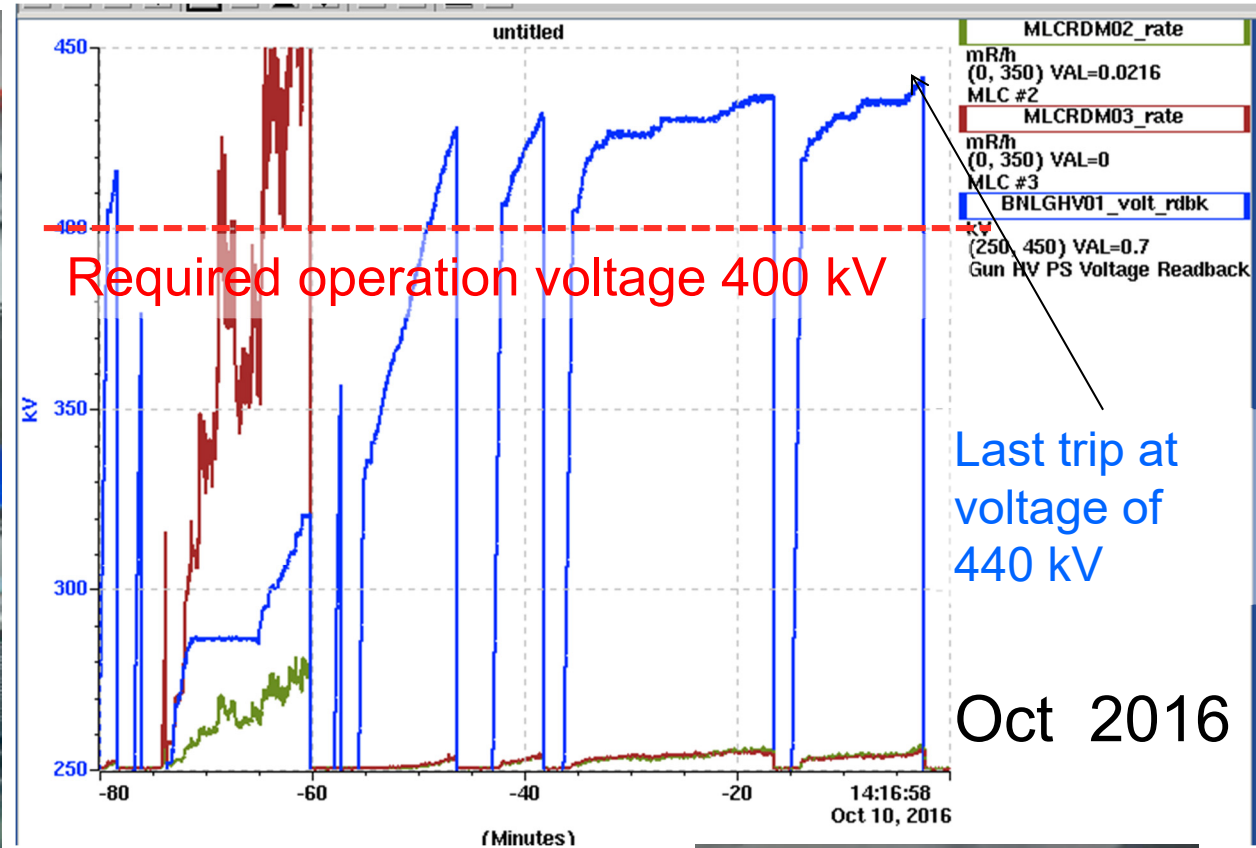
5



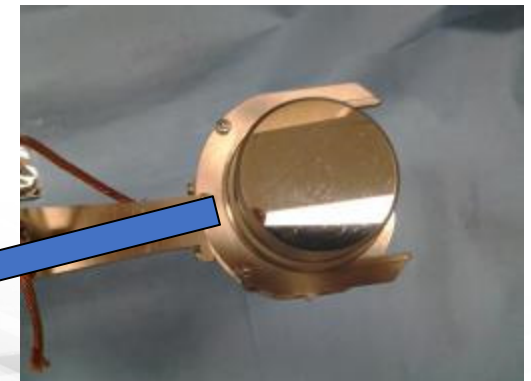
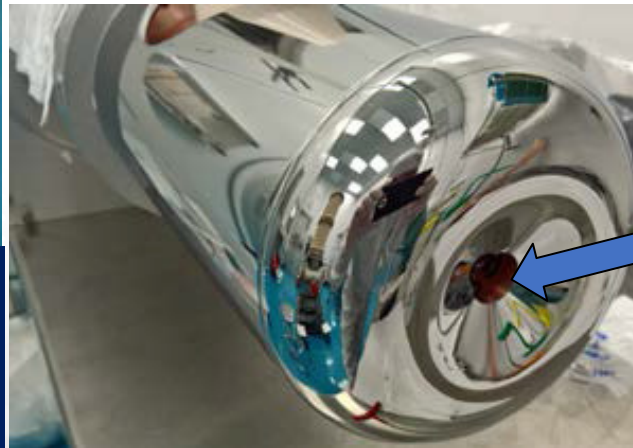
5



BNL DC gun performance during HV conditioning at Cornell

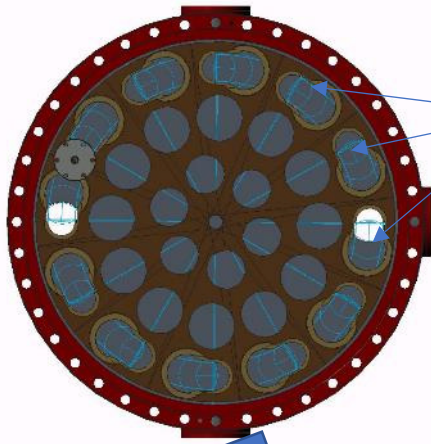


Designed, built and conditioned at Cornell University



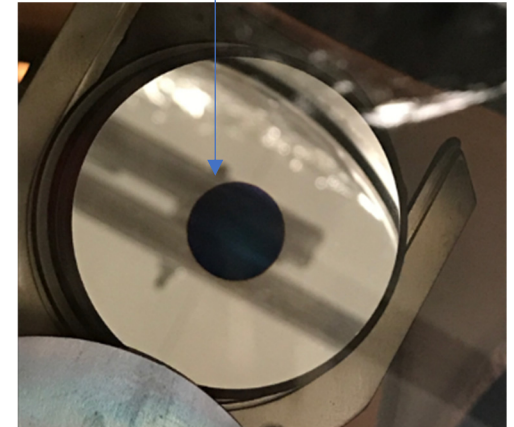
BROOKHAVEN NATIONAL LABORATORY
Molybdenum cathode plug

Photocathode production (in instrumentation building)

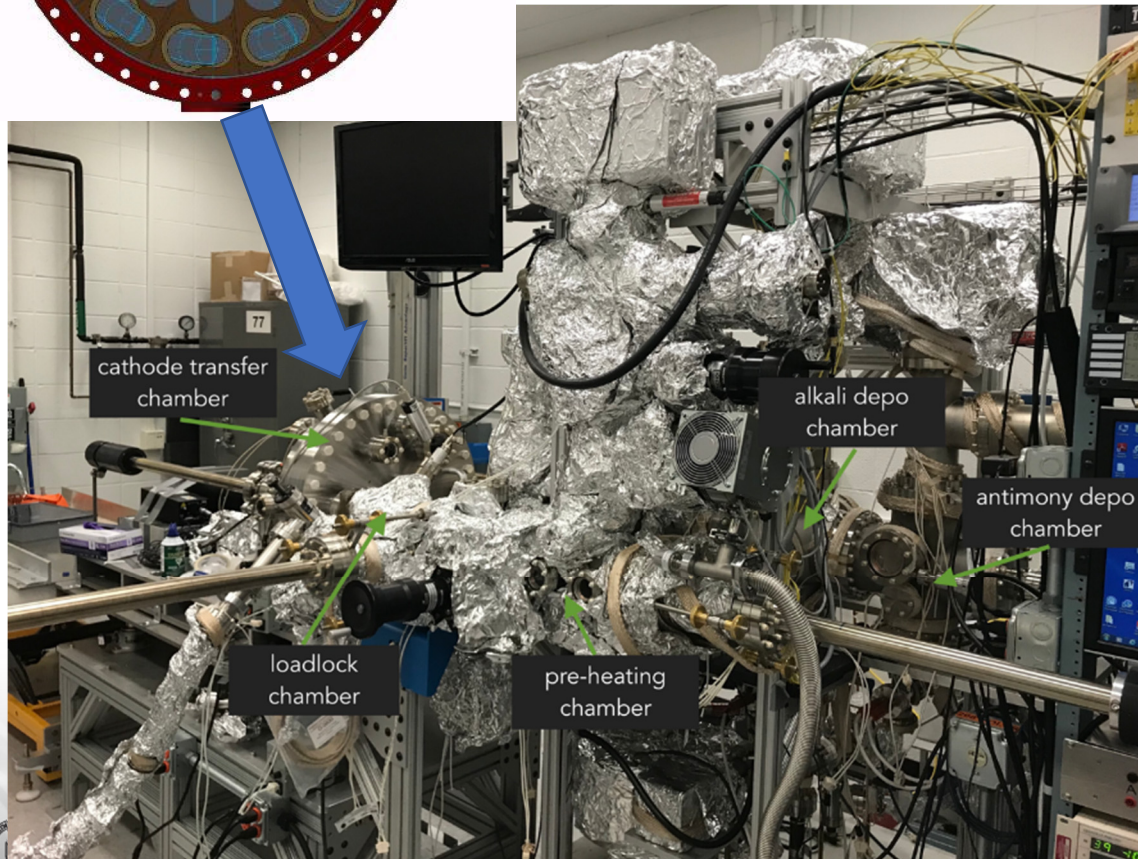
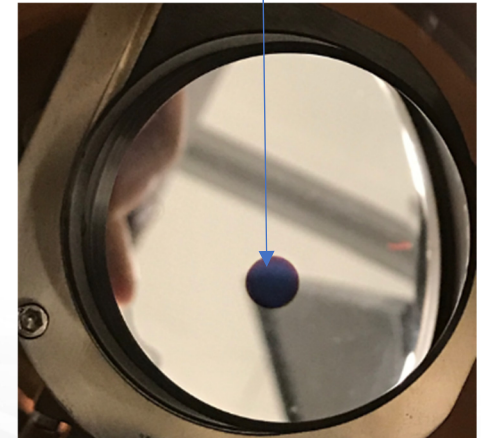


Cathode transfer camera
can hold up to 12 cathodes

On center 12mm active area
Used 2017-18 commissioning

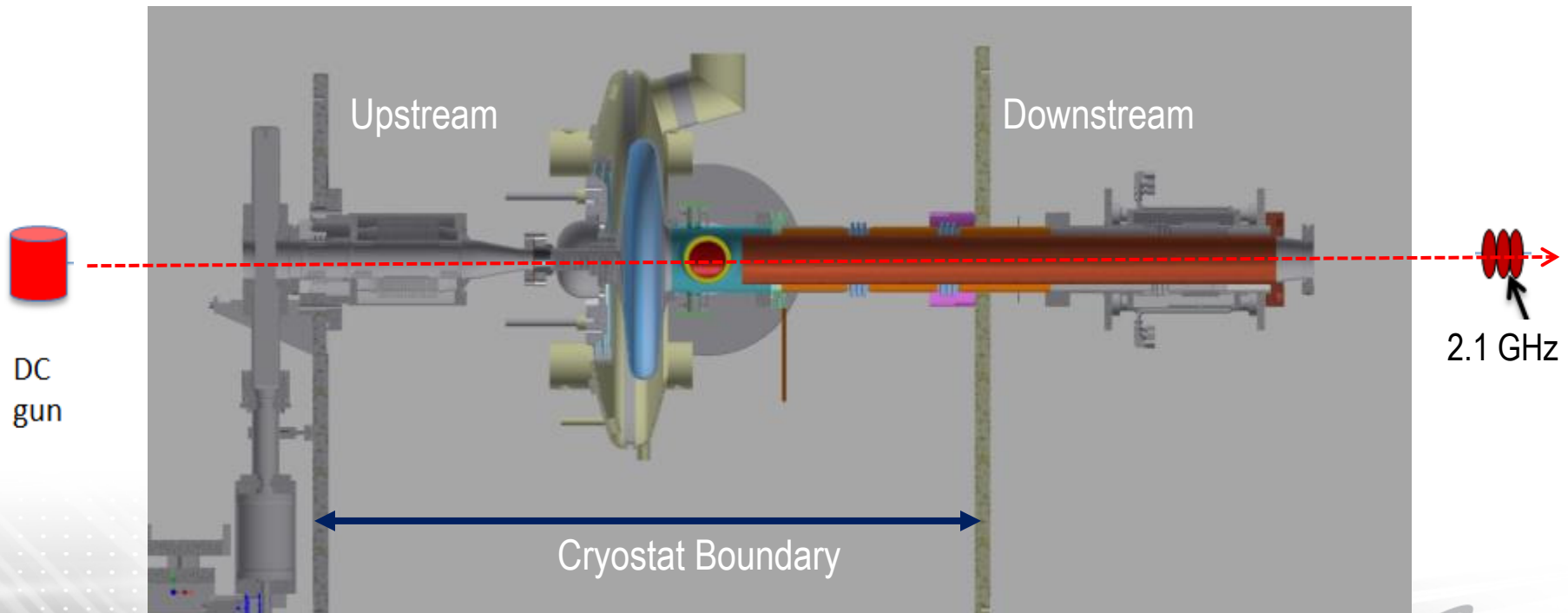


Off center 6mm active area
Used during 2019

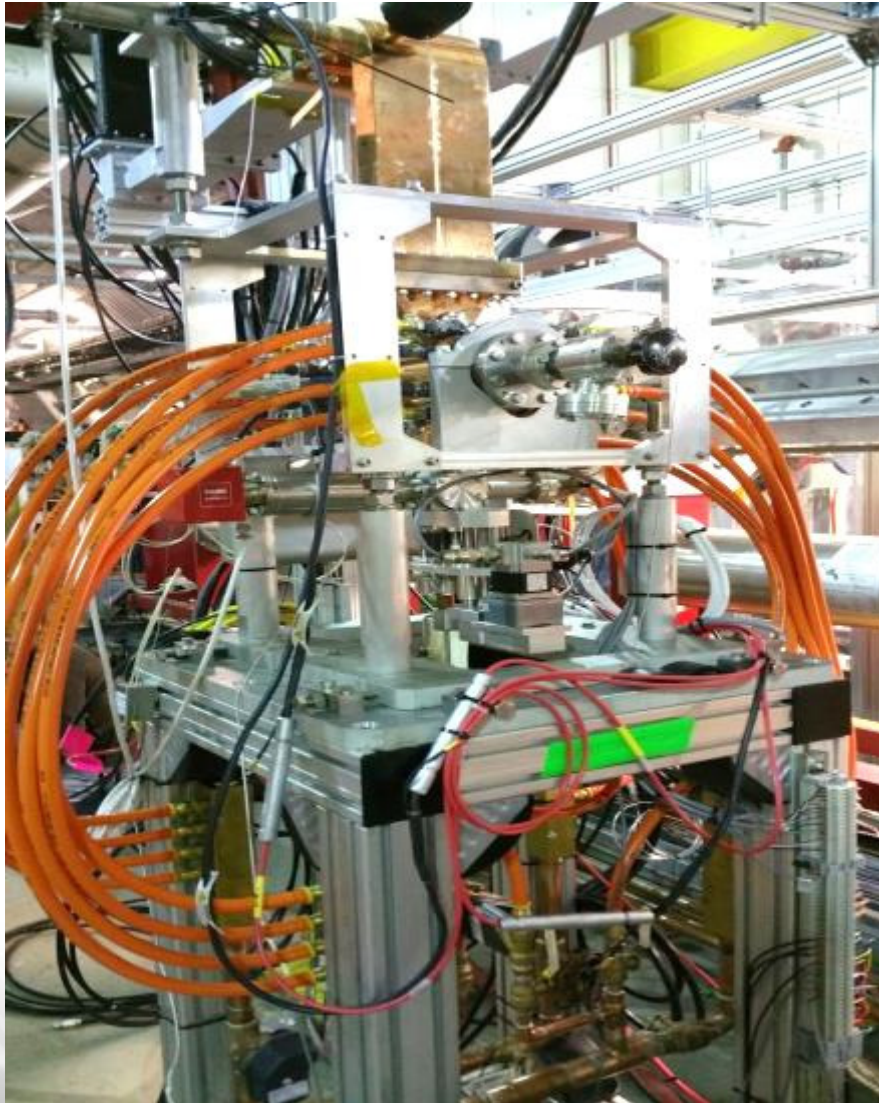


704 MHz SRF Booster Cavity (Modifications)

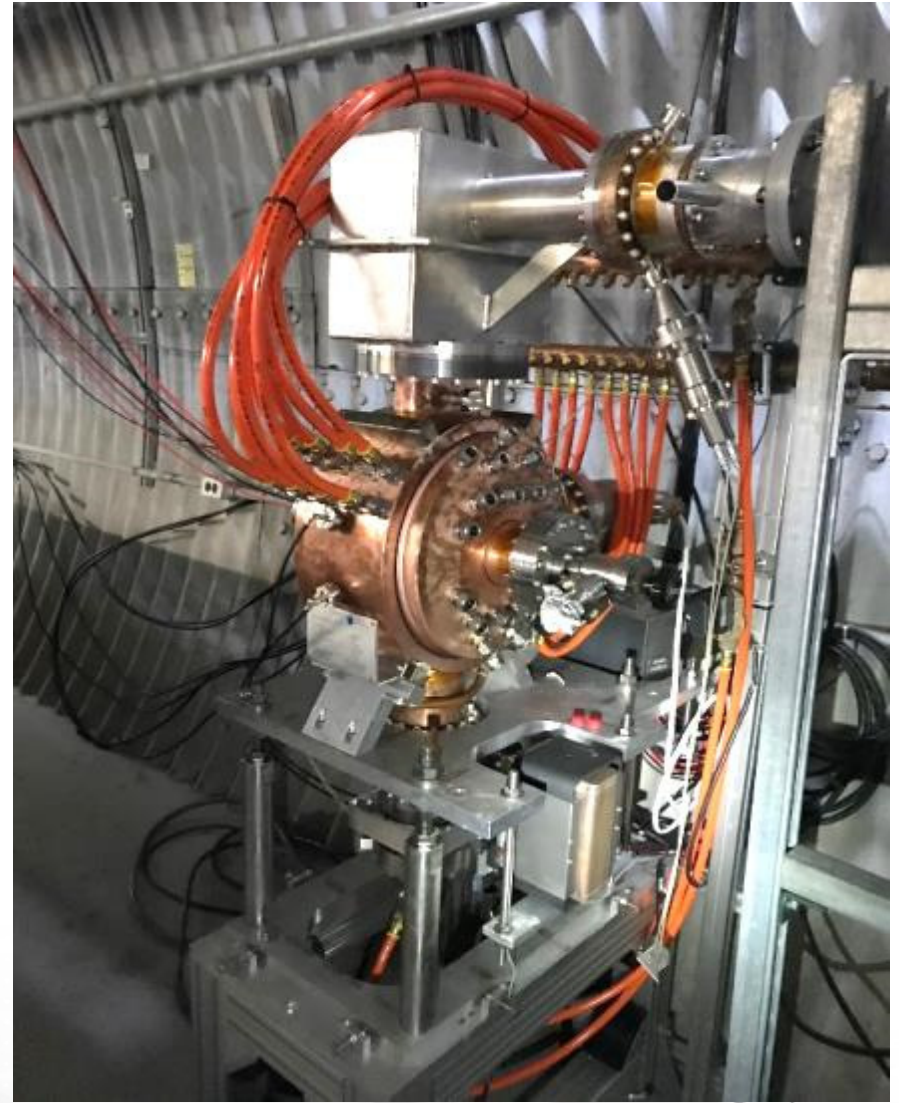
- Major modifications performed to convert from SRF Photocathode Gun to SRF Booster Cavity:
 - Upstream: Cavity modification to add new flange and “LHe Annex”. Upstream: Replace photocathode related components with new insert.
 - Downstream: Replace HOM damper with new insert and HOM absorber.
 - FPC: change the FPC penetration for $Q_{ext} = 1.7E5$, 120kW input power.
 - Cryomodule modifications to reduce microphonics.



2.1 GHz Cu Cavity (RF Curvature
Correction)



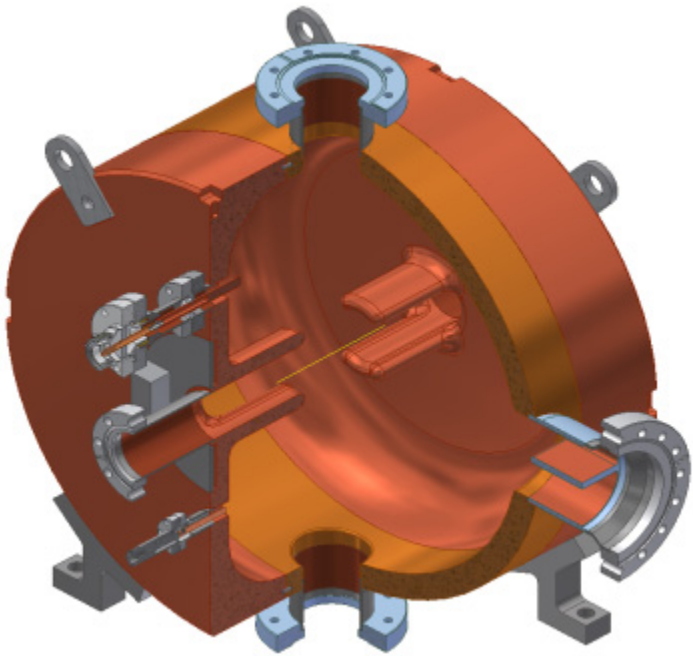
704 MHz Cu Cavity (De-Chirp)



9 MHz Cu Cavity (Transient Beam Loading Compensation)

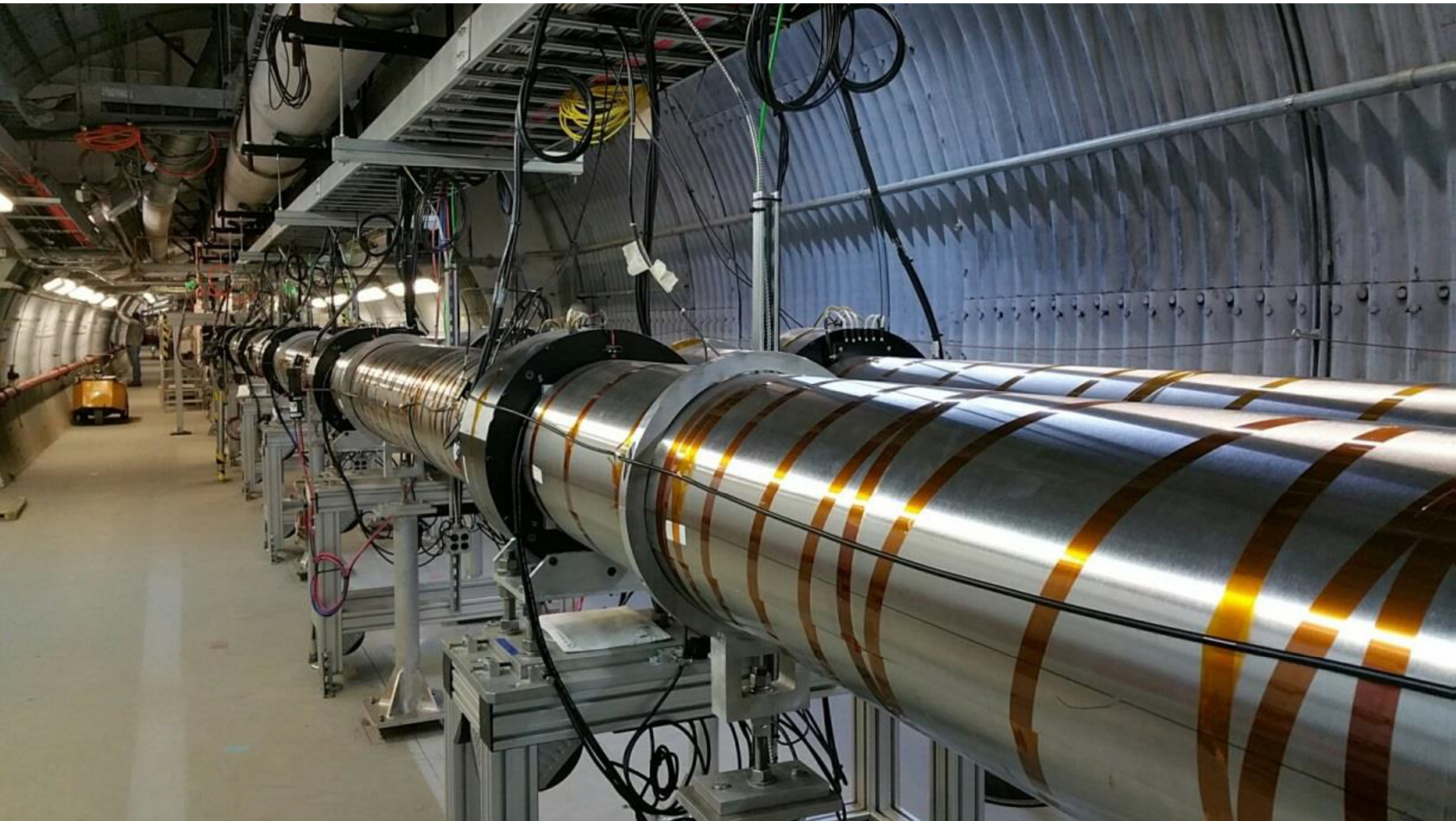


704 MHz Transverse Deflecting Cavity I

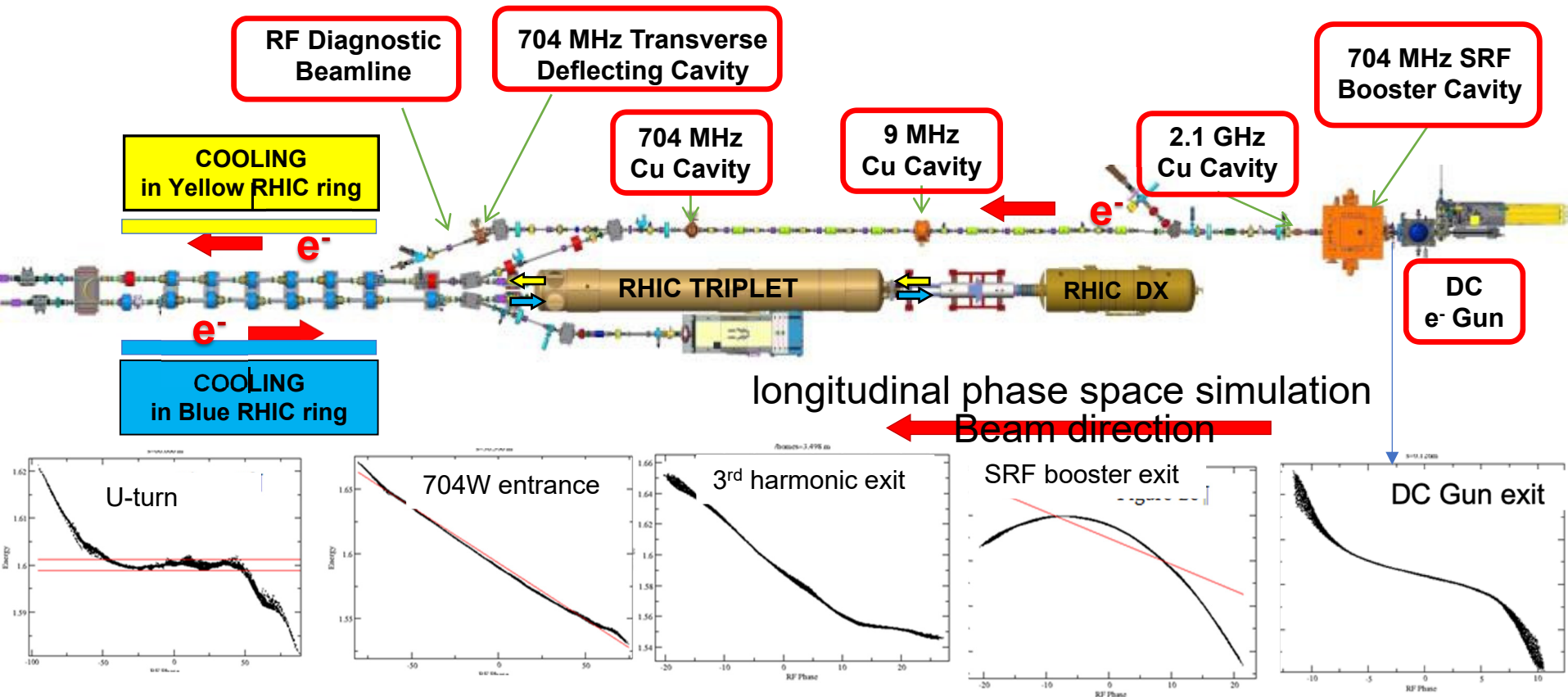


Design Value	$f=704.013$ MHz	$Q_{ext} = 13450$	$\frac{R_T}{Q}=1500$ Ω	$V=60.8$ kV
	f'	$ Q_{ext}' $	$ \frac{R_T}{Q}' $	$ V' $
Δx	0.170 MHz/mm	300/mm	$<2 \frac{\Omega}{mm}$	$<40 \frac{V}{mm}$
Δy	0.200 MHz/mm	100/mm	$<2 \frac{\Omega}{mm}$	$<40 \frac{V}{mm}$
$\Delta \phi_z$	0.080 MHz/degree	20/degree	$<3 \frac{\Omega}{degree}$	$<60 \frac{V}{degree}$

LEReC cooling section fully installed (2018)

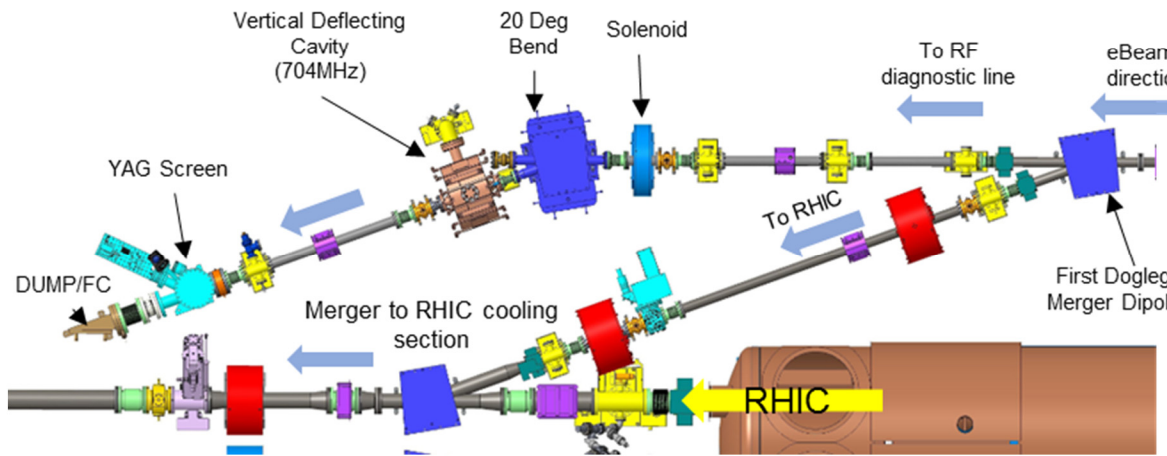


LEReC Accelerator Layout (RF Perspective)

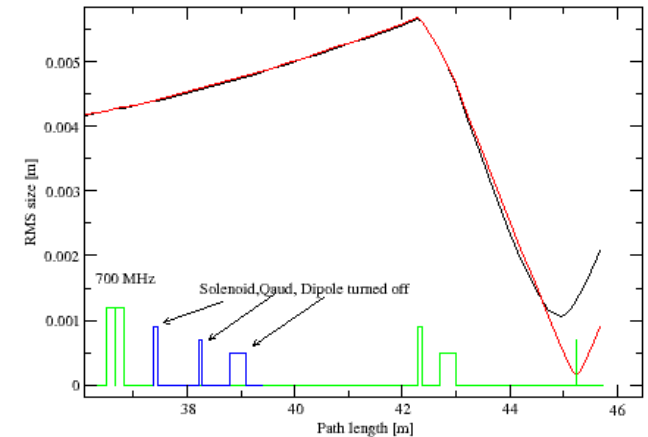


- 704 MHz SRF Booster Cavity = Acceleration, Energy Chirp for Ballistic Bunch Stretch
- 2.1 GHz Cu Cavity = RF Curvature Correction
- 704 MHz Cu Cavity = Removal of Energy Chirp
- 9 MHz Cu Cavity = Macrobunch Linear Transient Beam Loading Compensation
- 704 MHz Transverse Deflecting Cavity = Longitudinal Phase Space, Vertical Deflection to Provide Head to Tail Streak

RF diagnostic line for RF system fine tuning

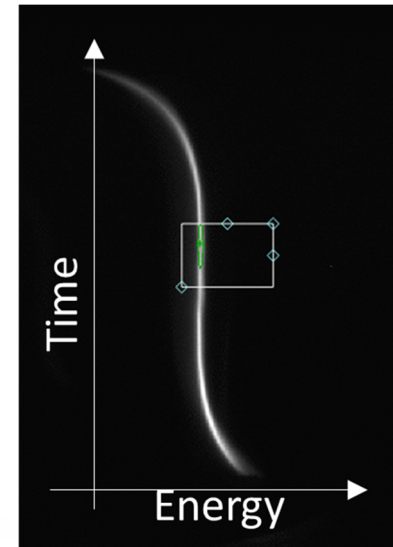


RMS envelopes in RF diagnostic line



If first dogleg merger dipole is turned off the electron beam is transported to the RF diagnostic line to measure longitudinal phase space profiles.

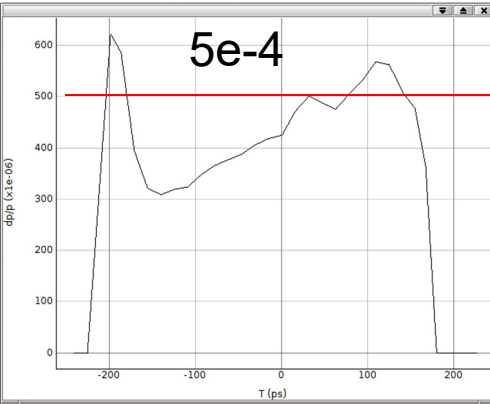
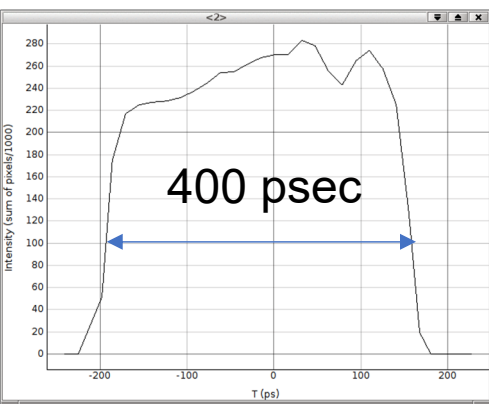
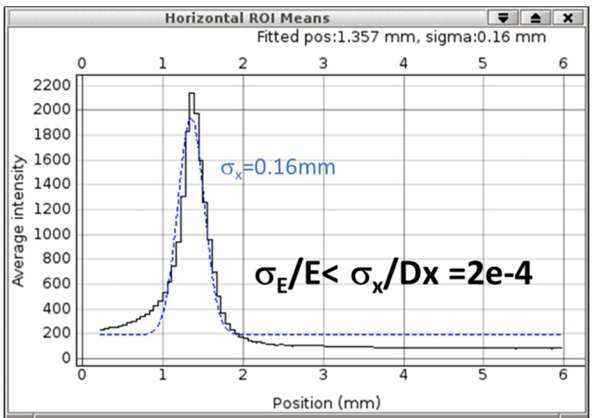
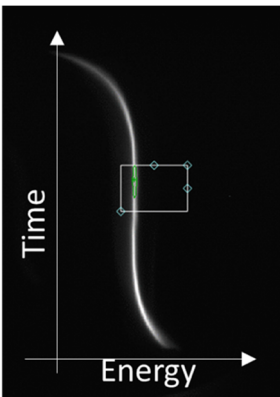
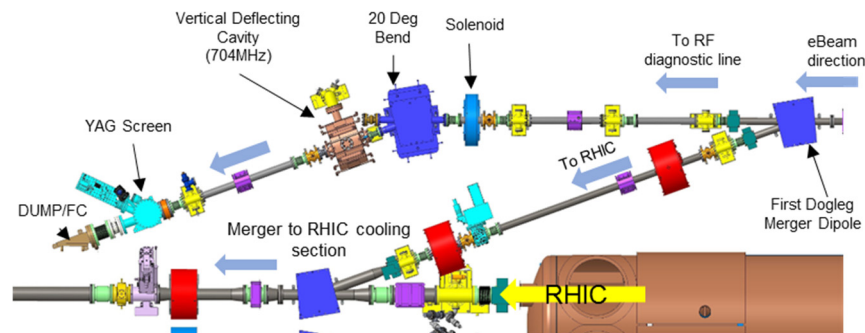
- 20 degrees dipole to generate a dispersion of 800 mm,
- solenoid provides small beta function at the YAG screen location,
- 704 MHz deflecting cavity to provide vertical time-dependent kicks.



Beam profile at
RF diagnostic
line YAG

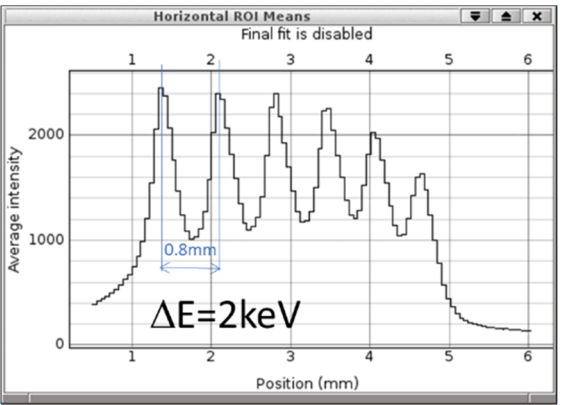
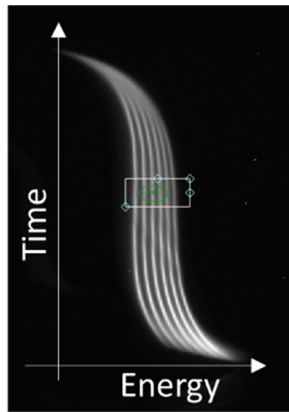
Longitudinal phase space measurement in RF diagnostic line

1 macrobunch, 3 nC



Longitudinal profile (left), and sliced energy spread (right) measurements at RF diagnostic YAG profile monitor for a bunch charge of 100 pC, FWHM of 400 psec.

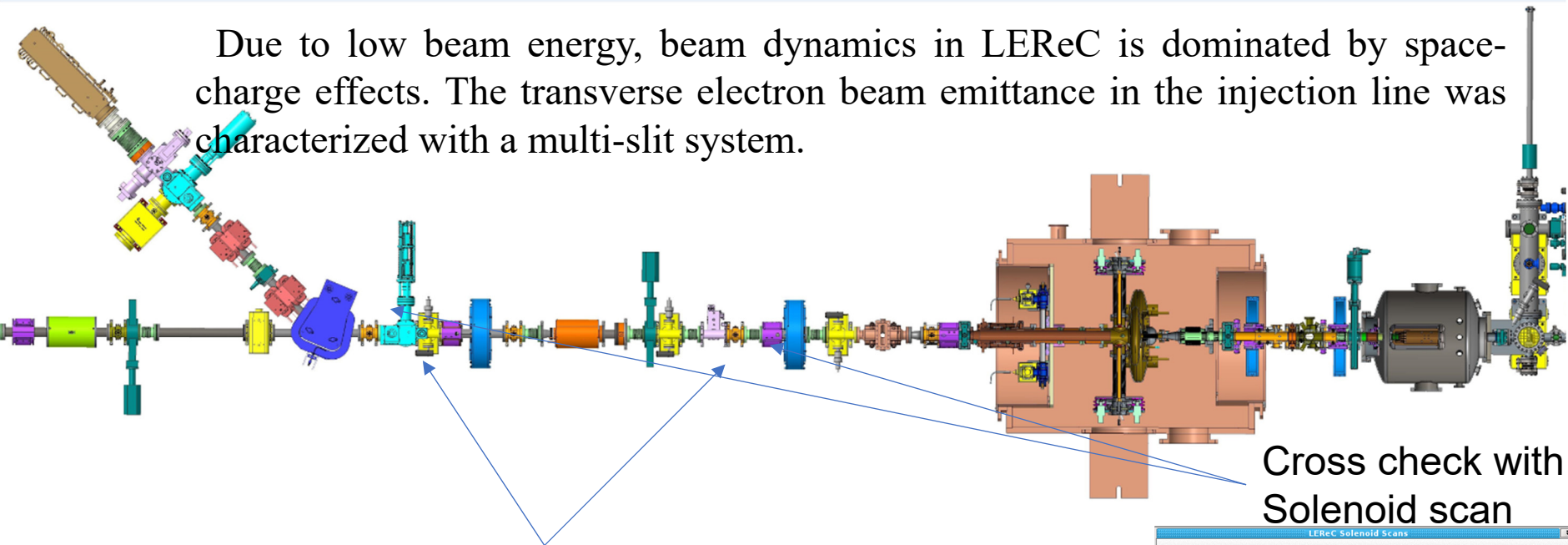
6 macrobunches, 3 nC each



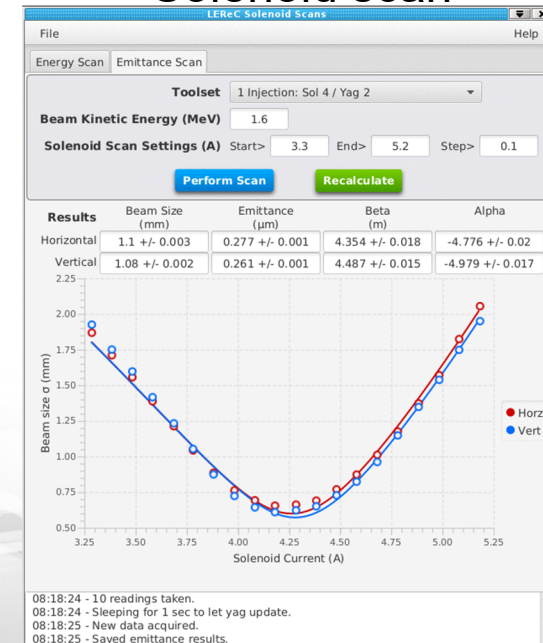
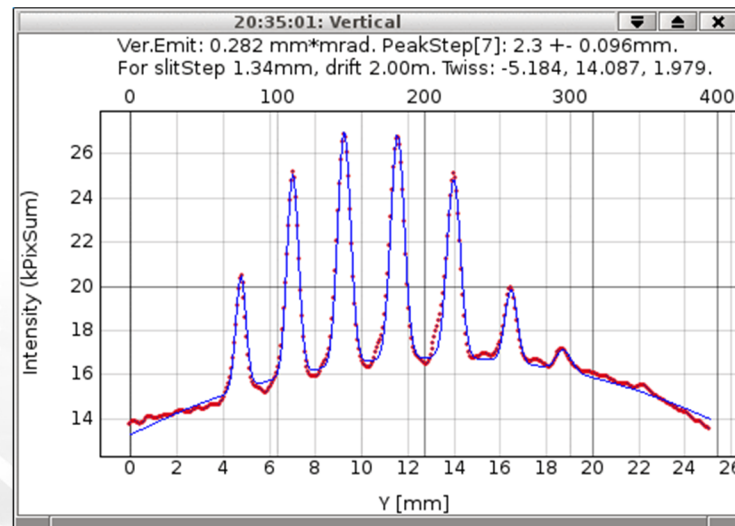
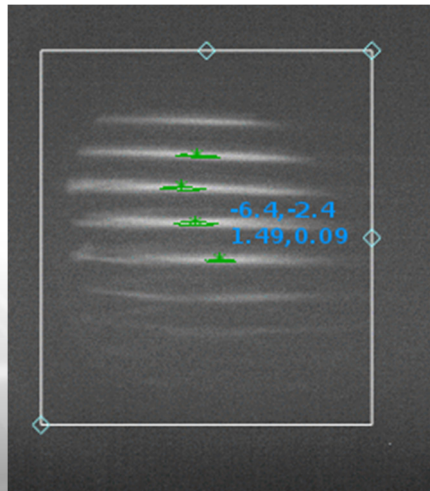
In pulsed mode due to beam loading effect followed bunches have lower energy

Transverse beam quality measurements

Due to low beam energy, beam dynamics in LEReC is dominated by space-charge effects. The transverse electron beam emittance in the injection line was characterized with a multi-slit system.



Multi slits emittance measurements



Transverse beam quality in cooling sections without ions

RF Diagnostic Beamline

COOLING
in Yellow RHIC ring

704 MHz Cu
Deflector Cavity

704 MHz
Cu Cavity

Transport
Beamline

9 MHz;
Cu Cav

Merger
Beamline

RHIC TRIPLET

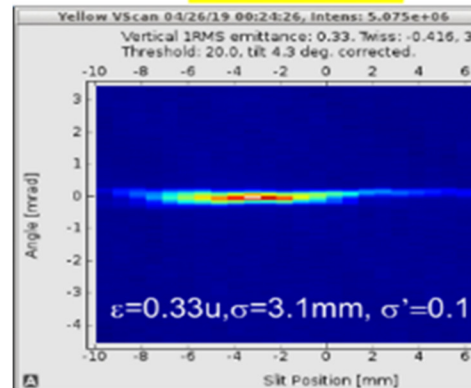
Extraction beam
YELLOW

High-Power Beam
Dump **BLUE**

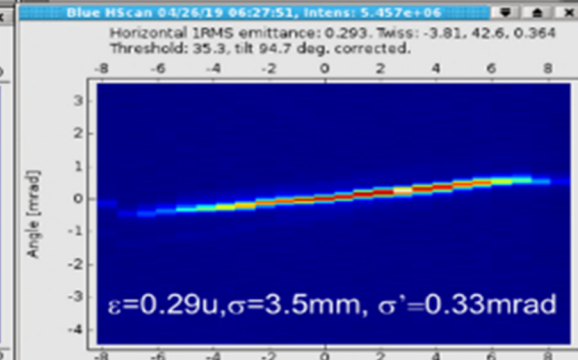
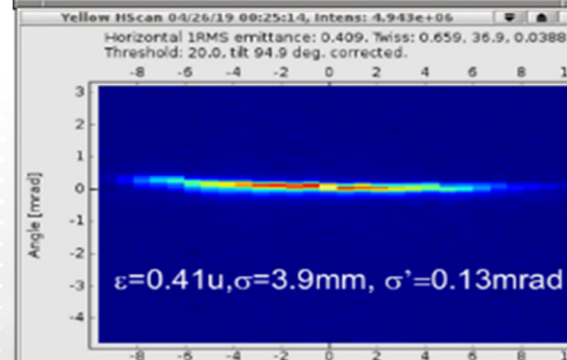
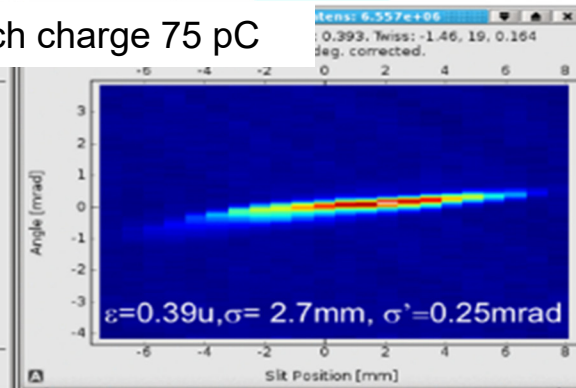
180°
Bending
Magnet

COOLING
in Blue RHIC ring

Movable slit and downstream
beam profile monitor are
installed at the beginning of
each of cooling section

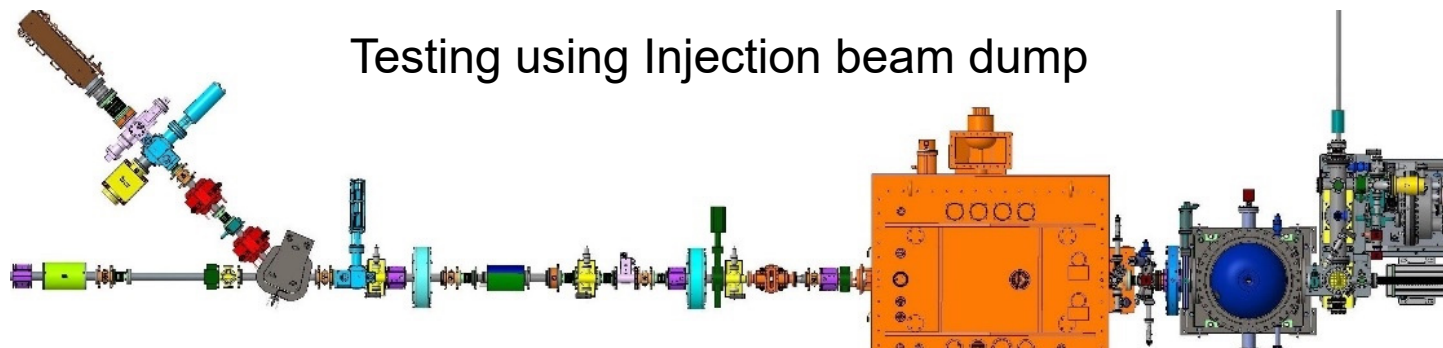


Bunch charge 75 pC



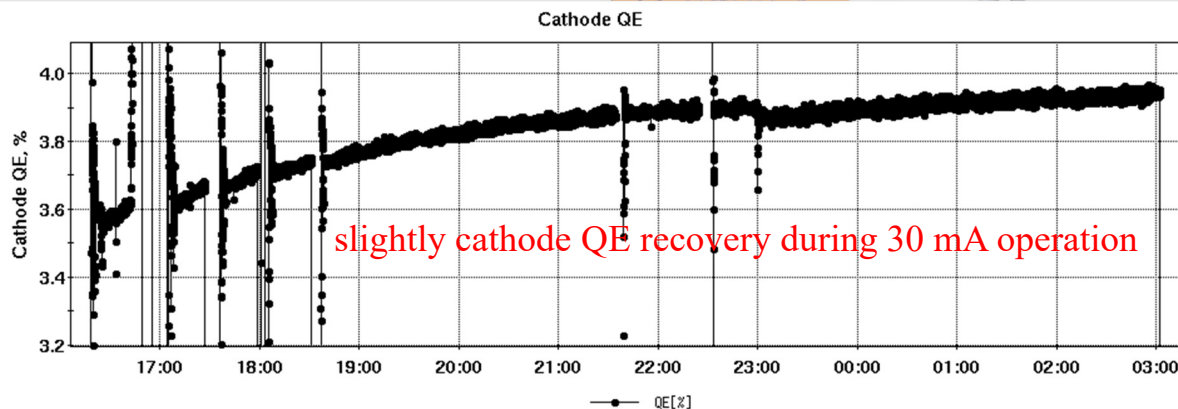
High current 9MHz CW operation to injection beam dump

Run 2018: stable 30 mA beams to the injection beam dump using reduced SRF booster voltage.

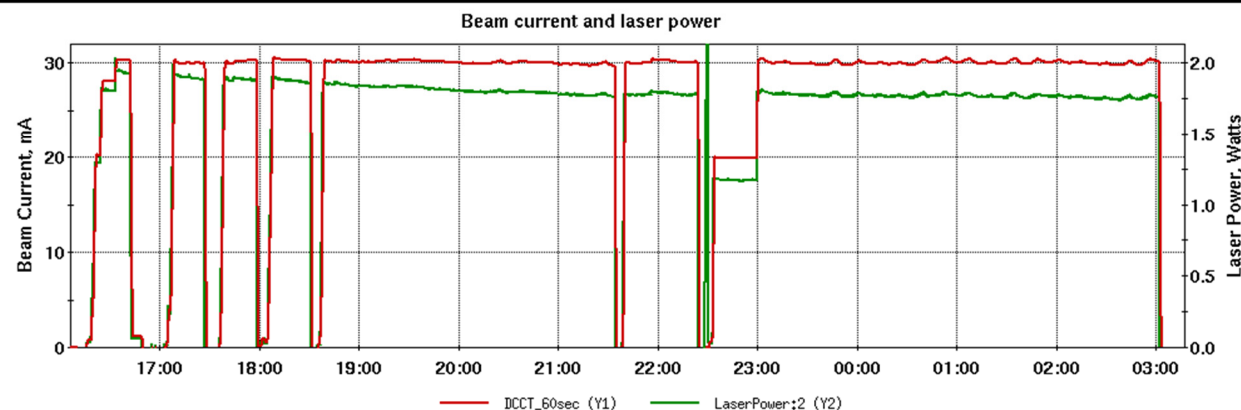


Testing using Injection beam dump

At the top plot:
cathode QE in %
(black).



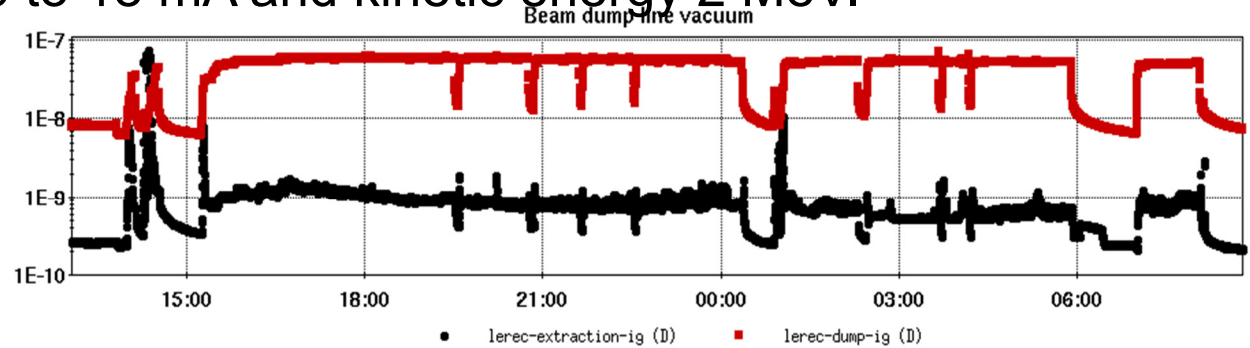
At the bottom plot:
laser average power
delivered to the
cathode (red) and
beam current
measured by DCCT
(green).



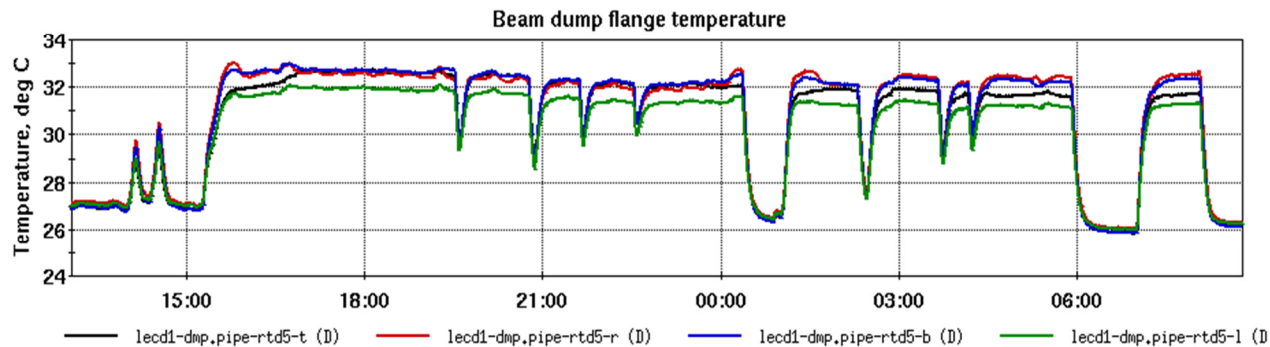
High current 9MHz CW operation full LEReC

operation to the final high-power beam dump at the end of run 2019 with average beam current up to 18 mA and kinetic energy 2 MeV.

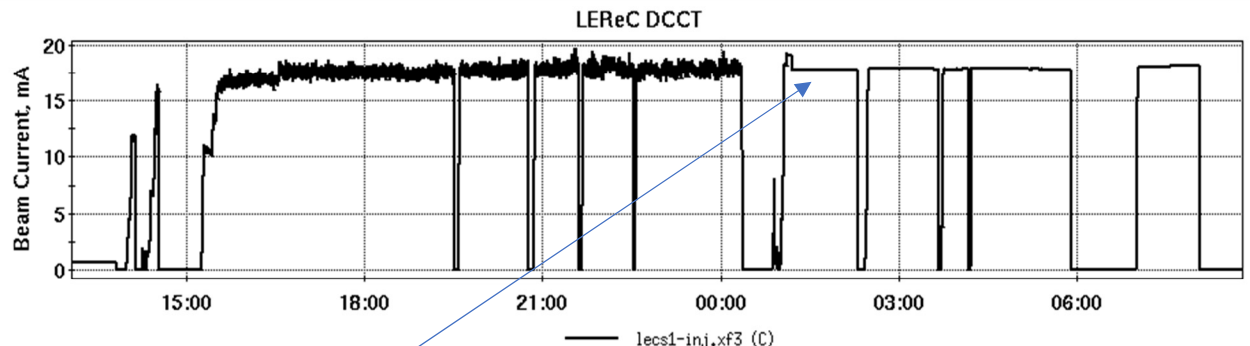
At the top plot, vacuum in the extraction line and near the beam dump,



at the middle plot beam dump entrance flange different sides temperatures,



at the bottom plot beam current measured by DCCT.

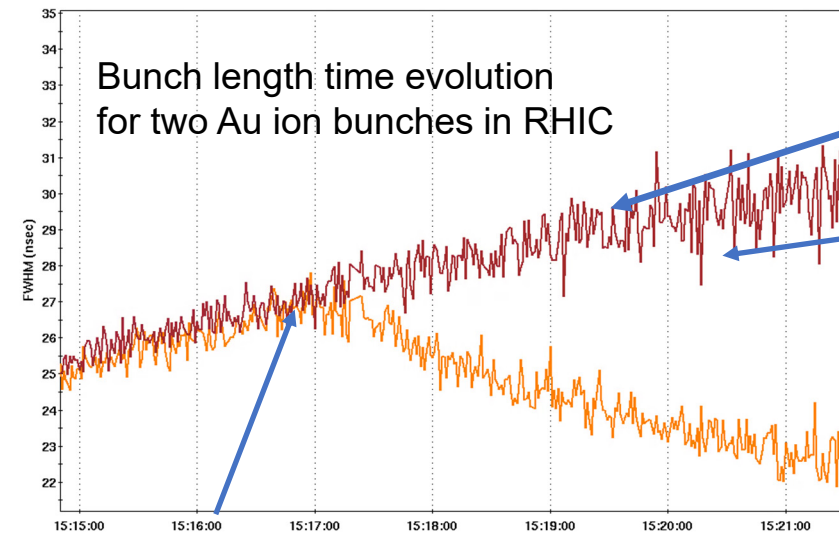


Intensity feedback was turned on at 01:00 am then beam fast intensity fluctuation has been reduced from 5×10^{-2} to 5×10^{-4} peak-to-peak.

LEReC: First observation of electron cooling using bunched electron beam (April 5, 2019)

Spring of 2019, a new RF timing system, including a 76 kHz mode of operation, was commissioned. The 76 kHz corresponds to the RHIC revolution frequency at gamma of 4.1.

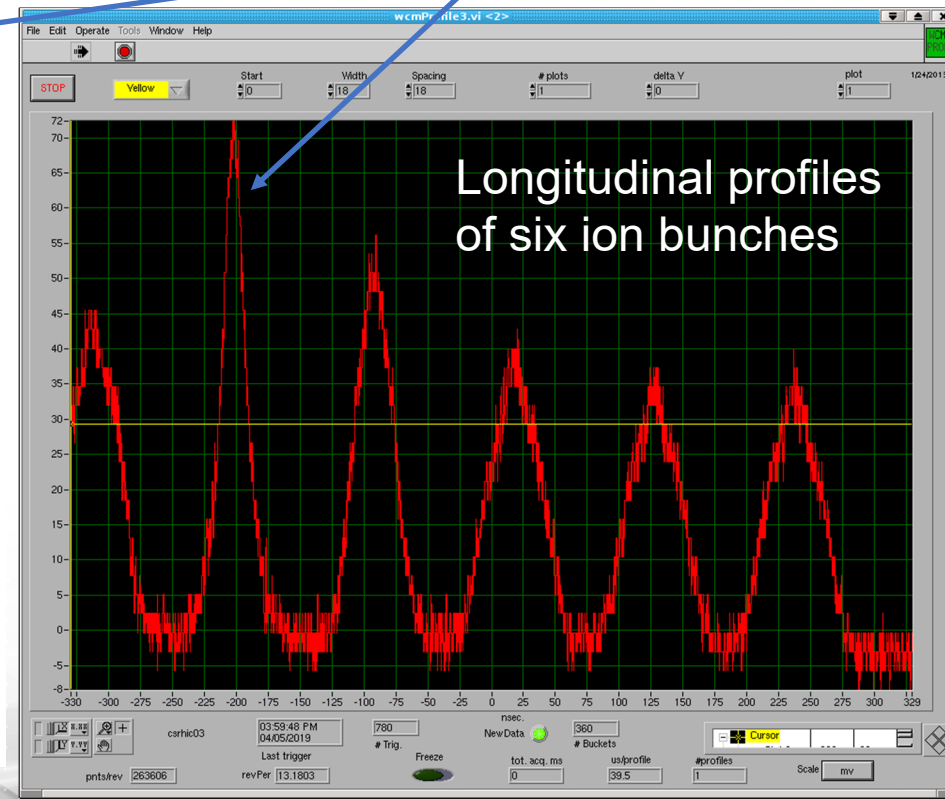
Cooling commissioning started with the 76 kHz mode of operation, which reduces average beam current requirement and average power by a factor of 120, while providing beam quality and interaction frequency sufficient to cool one ion bunch in each of the RHIC rings



Energy of electrons and ions matched

Ion bunch #4 which is not being cooled

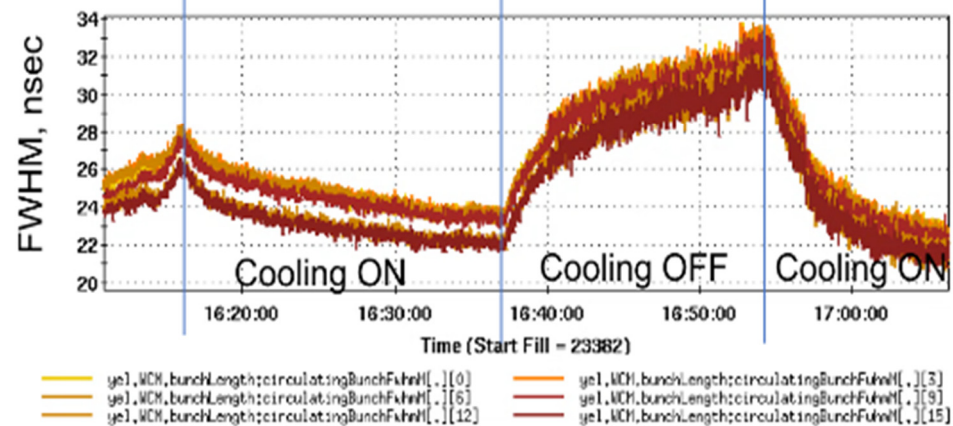
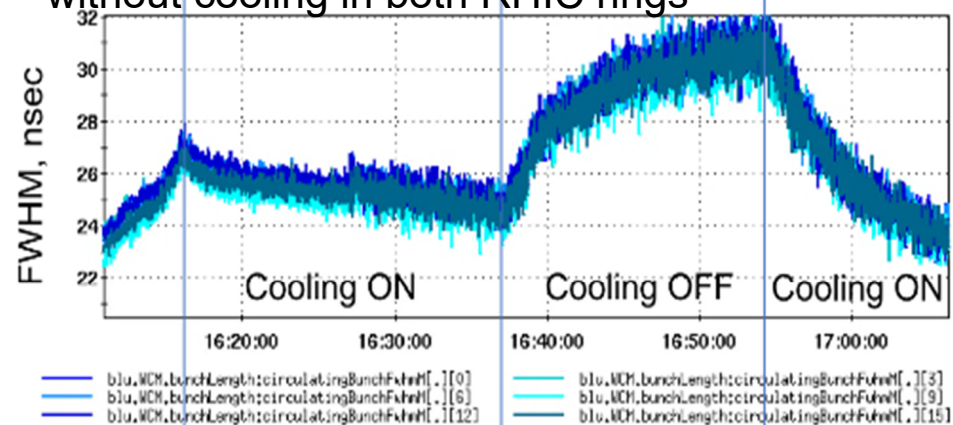
Ion bunch #2 is being cooled



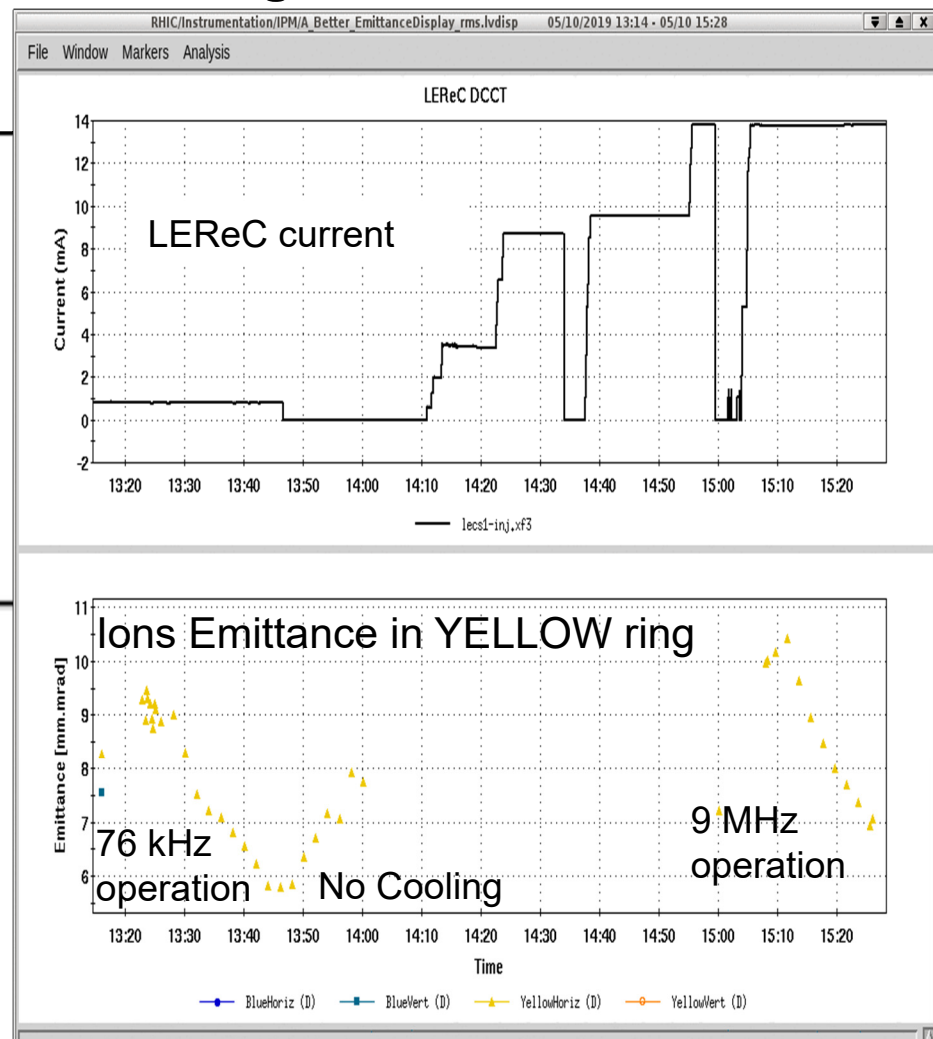
More about energy matching see talk today: TUZBB3 by S. Seletskiy "Precise beam energy matching for the experimental demonstration of ion cooling with a bunched electron beam."

CW operation at 14 mA and simultaneous cooling in Yellow and Blue RHIC rings, 6 bunches in each

Ion bunch length changes with and without cooling in both RHIC rings



Cooling in CW is not much different then in 76 kHz indicates that beam quality is satisfied LEReC requirement



More about cooling commissioning on Thursday : THZBA5 by A. Fedotov “First electron cooling of hadron beams using a bunched electron beam”

Summary and Plans

- LEReC is the first electron cooler based on the RF acceleration of electron beam.
- We designed, built and commissioned state of the art electron accelerator which provides beam quality suitable for electron cooling using bunched electron beams.
- First electron cooling using bunched electron beams based on the RF acceleration was demonstrated.
- Ion bunches in both RHIC rings have been cooled simultaneously using the same LEReC beam.
- With a bunched beam electron cooling technique now experimentally demonstrated, its application to high-energy cooling can open new possibilities by producing high-quality hadron beams
- The next step, optimize of LEReC cooling efficiency and effects on ion beam lifetime to maximize collision rates during the next RHIC low-energy run in 2020.

List of LEReC relative presentations at NAPAC2019

- **MOYBA6.** Chuyu Liu et al., Accelerator performance during the Beam Energy Scan II in 2019.
- **MOPLM18.** P. Inacker-Mix et al., Design of the 2-stage Laser transport for the Low Energy RHIC electron Cooling DC Photogun.
- **MOPLO11.** Honghai Song et al., Recent Advances in Rotating Coil and Hall Probe Measurements at BNL
- **TUZBB3** S. Seletskiy et al., Precise beam energy matching for the experimental demonstration of ion cooling with a bunched electron beam.
- **THZBA5** A. Fedotov et al., First electron cooling of hadron beams using a bunched electron beam.
- **THYBA6, MOPLM22.** L.Nguyen et al., Active pointing stabilization techniques applied to the Low Energy RHIC electron Cooling Laser transport.

Acknowledgement

LEReC success would not be possible without team effort and expertise of many people from various groups of the Collider-Accelerator and other Departments of the BNL.

As well as FNAL, ANL, JLAB and Cornell University.

Thank you!



LEReC team